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12.6% Fewer Vehicles, Zero Regressions: An Audit-First ALNS for Electric-Vehicle Routing

Yiyang Guo | Department of Mathematical Sciences

Motivation



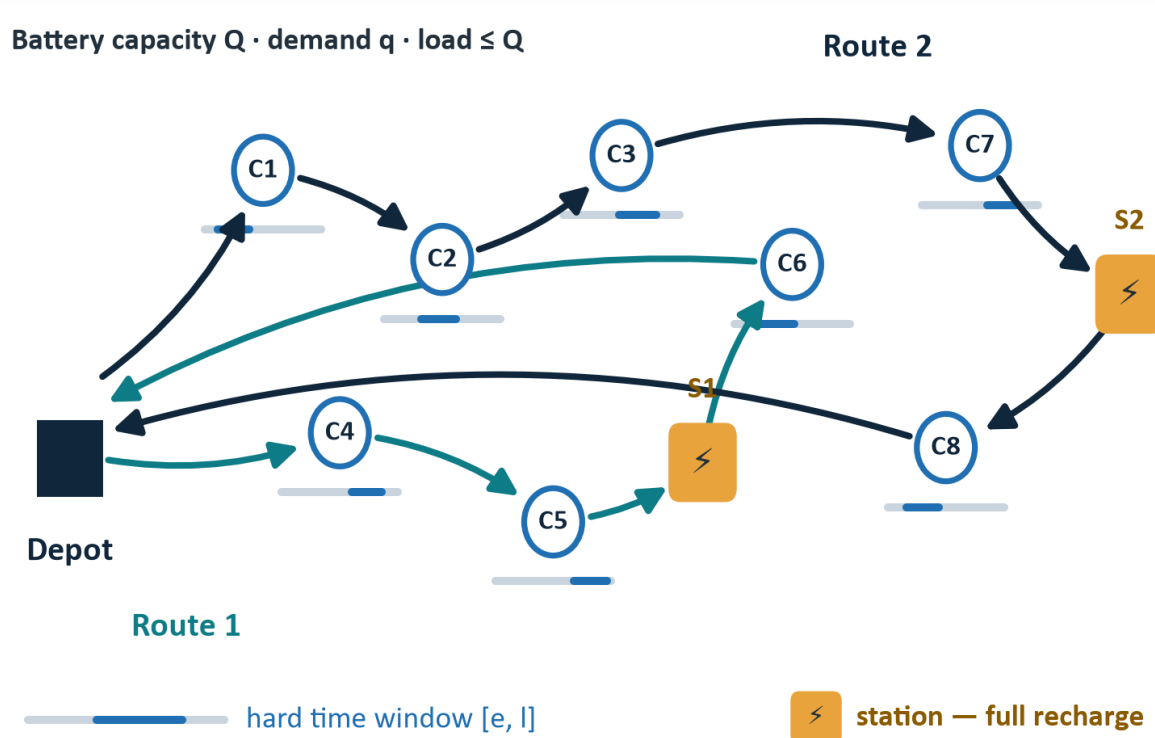
E-VRPTW: battery-limited EVs, hard time windows



Fleet size first in the lexicographic objective



Feasibility hinges on exact charging decisions



Objective — reproduce SSG (2014), then cut fleet size with zero regressions

Benchmark — SSG (2014) · 12 instances · 5–100 customers · C / R / RC

Model — battery-limited EVs · en-route charging · hard time windows

Toolchain — Python · numba · CPLEX · Gurobi · HIGHS · OR-Tools (lockfile-pinned)

Method Overview

INITIAL SOLUTION

greedy insertion · one route per seed customer cluster

DESTROY

constraint-guided removal:
station pressure · TW conflict · energy detour

REPAIR

vehicle-count-aware greedy insertion
(existing routes first, new route last)

PREFILTER

capacity · optimistic time · optimistic energy
(cheap, shared, screens before exact work)

EXACT CHARGING

labeling subproblem,
solved per changed route only

ACCEPT?

simulated annealing —
never accepts more vehicles

lexicographic objective: vehicles $\gg$ distance $\gg$ charging time

Constraint-guided ALNS engine — vehicle-first search with exact charging subproblems

Exact charging dominates cost: 1,687–2,161 calls per 30 s run

Acceleration shipped: incremental cache · exact-deadline · parallel control

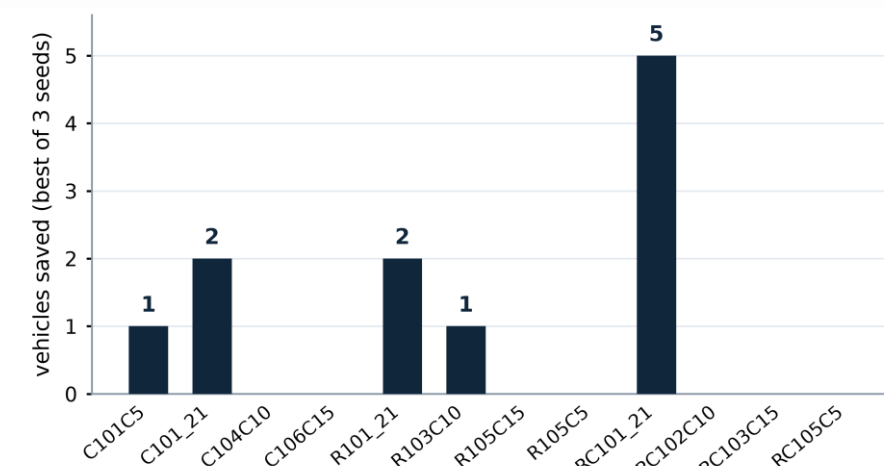
KEY MECHANISMS

- Vehicle-count-aware repair: eliminate routes first; fleet can fall, never rise
- Optimistic prefilters (capacity · time · energy) screen before exact evaluation
- Constraint-guided destroy: station pressure, TW conflicts, energy detours
- Exact charging subproblems re-solve only the routes that changed

AUDIT PROTOCOL

- Lexicographic objective — vehicles first; acceptance rejects any fleet-size increase
- Independent review CLI replays solutions, event logs & hashes before publication

VEHICLE SAVINGS



Best of 3 seeds per instance — total fleet 87 $\rightarrow$ 76 vehicles (–12.6%)

Experimental Settings

every stage gated by an independent CLI review that replays raw JSON + event logs

- Stage 0**
Baseline reproduction of SSG (2014) — 36/36 feasible, frozen forever
- Stage 1**
Lexicographic objective: vehicles $\gg$ distance $\gg$ charging time
- Stage 2.1–2.3**
Route reduction $\rightarrow$ constraint-guided ALNS
–12.6% vehicles · –5.1% distance · zero regressions
- Stage 3.0–3.4**
Cheap screening · cache · exact deadline · control parallel
–22% median exact-charging CPU (C set)
- Stage 4**
Adaptive operator weights — wins on 7/36 runs (144/144 re-verified)
- Stage 5.1**
Honest BKS audit — 92/124 entries model-incompatible, no fake gaps

Instances

12

Seeds

3

Time limit

30 s

Iterations

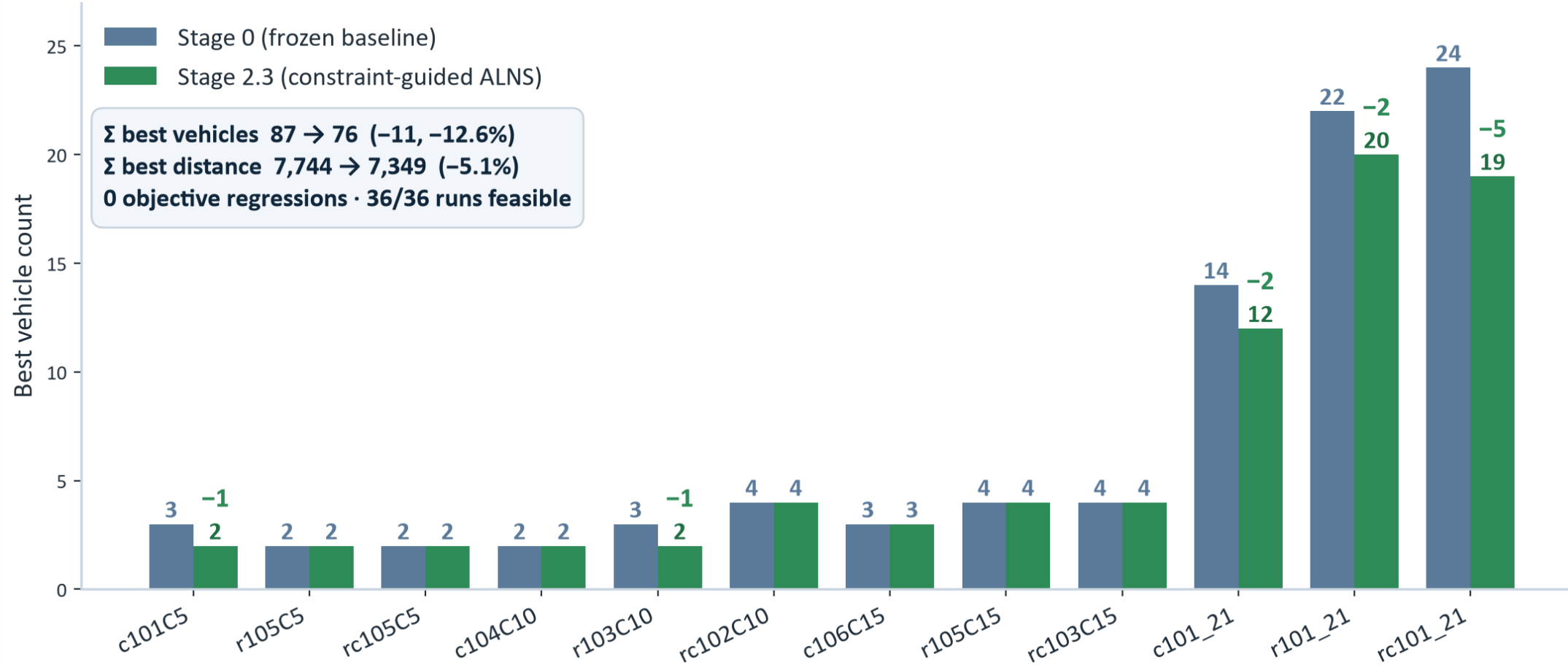
1,000

Thread

1

Stage 0 frozen & hash-verified — every number replays from raw evidence

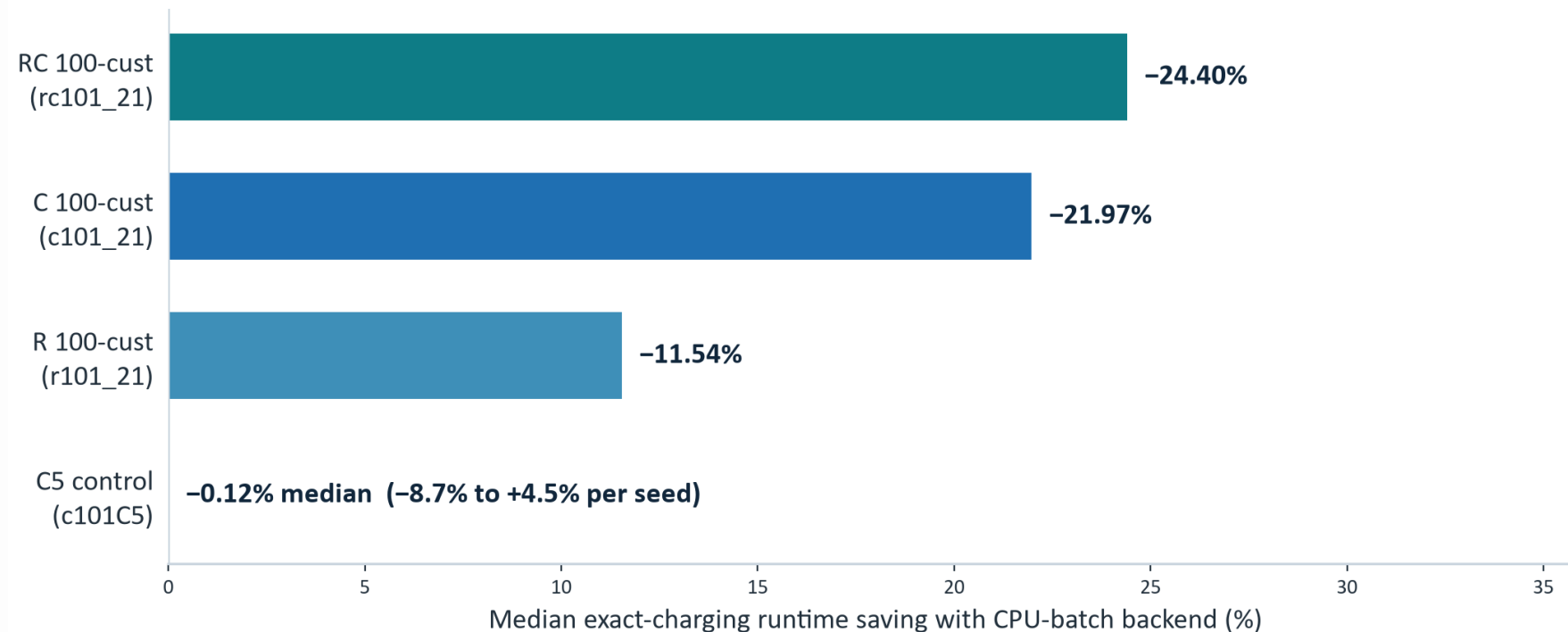
Key Results



Vehicle-first search cuts fleet on 5/12 instances — 87 $\rightarrow$ 76 vehicles, –5.1% distance, 36/36 feasible

Per-family best — C 22 $\rightarrow$ 19 · R 31 $\rightarrow$ 28 · RC 34 $\rightarrow$ 29 · mean fleet 90.7 $\rightarrow$ 77.0

Key Quantitative Highlights



–22.0% exact-charging CPU (median, C instances) — CPU-batch default after audit

Batch pilot covers 4 instances (speedups 1.00–1.32x) — full-scope acceleration remains a Stage 3 target

Why It Works & Honest Audit

✓WHY IT WORKS

- Vehicle-count-aware repair — fleet can fall, never rise
- Constraint-guided destroy targets station pressure · TW conflicts · energy detours
- Dynamic removal size escalates on stagnation (17,892 logged)
- Stage-gated: two independent complete reruns must agree

▲HONEST BOUNDARIES

- 92/124 published BKS entries are model-incompatible — no gap claims against them
- C5 control: –0.12% median (–8.7%...+4.5%) — speedup is structure-dependent, not universal
- 9/36 runs overran the 30 s budget by ≤ 0.02 s — all kept, with events
- No unattested baselines — every number replays from raw solution JSON + event logs

EVIDENCE AT SCALE

12,121

evidence files on disk

200,731

operator event rows replayed

20

real failure cases kept

0

synthetic cases in audit

[1] Schneider, Stenger & Goeke (2014) Transp. Sci. 48(4):500–520 [2] Pisinger & Ropke (2006) Transp. Res. B
[3] Keskin & Çatay (2016) Transp. Res. E [4] Hottung, Kwon & Tierney (2022) EJOR



Scan for code & evidence



–5.1% total distance



36/36 feasible formal runs



–22.0% exact-charging CPU